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Directorate A – Digital Transformation
DIGIT.A.3 – Solutions for Legislation, Policy & HR



Document templates and elements configuration

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1. Introduction

LEOS (Legislation Editing Open Software) is a project under [Interoperable Europe](#) initiative of the European Commission for a reinforced public sector interoperability policy, funded by the [Digital Europe Programme](#) (DIGITAL) and which was created to address the need of the public administration and European Institutions to generate draft legislation in a legal XML format.

The LEOS project under **Interoperable Europe** focuses on supporting the co-development co-design, and co-deployment of an 'IT eco-system centred on augmented LEOS'.

LEOS has been created to addresses the **modernisation and digital transformation of the drafting and revising of legislation in EU Institutions, EU agencies and bodies and Member States**.

LEOS ensures that the content drafted by the users follows the drafting guidelines by offering features like enforcing predefined document structures, predefined layout, numbering rules. All of this to ensure that the author would first and foremost focus on drafting itself and much less on layout management (or checking). In order to facilitate efficient online collaboration LEOS also contains some others features like comments, suggestions, version control, co-edition, etc.

Content is stored in an XML format, currently in AKN4EU which is based on the Akoma Ntoso OASIS standard namespace: <http://docs.oasis-open.org/legaldocml/ns/akn/3.0>. This facilitates document exchange, element retrieval, etc.

While Akoma Ntoso defines representations for parliamentary, legislative and judiciary documents, AKN4EU is a localisation of Akoma Ntoso and currently focus on the documents exchanged in the scope of the Ordinary Legislative Procedure (OLP).

- Regulation, Directive, Decision,
- Proposal for a Regulation, Proposal for a Directive, Proposal for a Decision.
- Some cases of the EP Texts Adopted.
- Council documents including the Position and the Statements of Council reasons.

(see AKN4EU 4.0 Baseline – General structure of documents, 0 Scope).

Each of the above legal documents have a certain structure, and it is usually composed of a cover page and one or more components. The components have their own structure.

Therefore, the main concept of LEOS is managing the documents (the legal documents) and their components (or subdocuments).

LEOS is delivered with a set of [templates](#) and offers some elements to allow organizations to define and configure all these things according to their needs. This is explained in this document.

2. Prerequisites

Install LEOS

The current modules of augmented LEOS are:

- [LEOS core](#) and
- [Annotations utils](#).

IMPORTANT NOTES: *This release is intended to provide an experience with the software and is stripped of several important components to enable ease of use.*

Additionally, this software is adapted to run on a local server for demo purposes and without proper security mechanisms.

DEPENDENCIES

To compile the supplied source files and run the generated WAR the following software should be configured:

- Java SDK version 8.0.
- Maven version 3.3.9+ (Maven runtime memory might need to be set: MAVEN_OPTS=-Xms256m -Xmx512m) (Maven settings, proxy and mirrors, might need to be adjusted to your environment and internet access requirements).
- Supported browser is Google Chrome version 45+ (Mozilla Firefox ESR version 38.3 and Microsoft Internet Explorer version 11 are known to work with minor issues).
- Node 18.18.2.

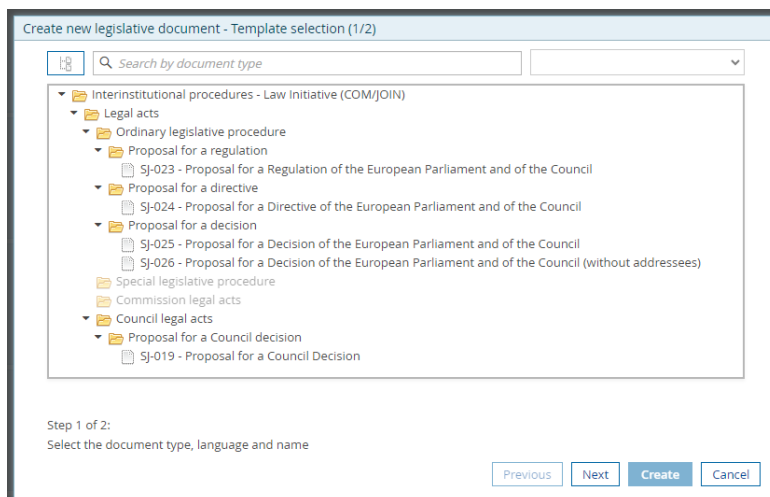
For further instructions on how to install LEOS , please consult the [Getting started](#) guide.

3. Document templates and element configuration

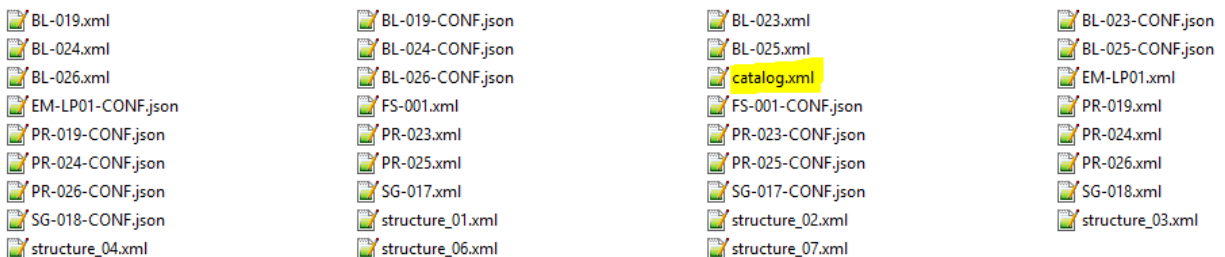
Document templates are XML documents, defined in format Akoma Ntoso and are stored in the document repository. They are used for the instantiation of the components/subdocuments comprised in the Act/Package.

In LEOS application, a user can select the type of document to be created by clicking the "**Create proposal**" button.

The term "act" is the type of document that is currently used by default in LEOS, but that can be changed, depending on the specific needs.



All the information needed to compose the template selection screen and the documents that belongs to each of the acts are store in the "**catalog.xml**", located in {AUGMENTED_LEOS}/leos/tools/repository/server/src/main/resources/leos/templates/consolidated/os



When adding or removing template(s), the "**catalog.xml**" needs to be updated accordingly.

The catalog.xml is the entry point to describe all existing templates.

For example, proposal "SJ-023 - Proposal for a Regulation of the European Parliament and of the Council" is composed of documents and subdocuments:

- Proposal document template **PR-023**,
- Memorandum (sub)document template **EM-LP01**. (*The Memorandum document template can be removed and then no memorandum will be created.*),
- Legal Text (sub)document template **BL-023**,
- Annexes "**SG-017**" and
- Financial statement "**FS-001**".

```
<item type="CATEGORY" id="cl.1.1.1" enabled="true" key="REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL">
  <names>
    <name lang="EN">Proposal for a regulation</name>
    <name lang="FR">Proposition de règlement</name>
  </names>
  <descriptions>
    <description lang="EN">Regulation to be adopted by the European Parliament and the Council</description>
    <description lang="FR">Règlement à adopter conjointement par le Parlement européen et le Conseil</description>
  </descriptions>
</item>

<item type="TEMPLATE" enabled="true" key="SJ-023">
  <names>
    <name lang="EN">SJ-023 - Proposal for a Regulation of the European Parliament and of the Council</name>
    <name lang="FR">SJ-023 - Proposition de règlement du Parlement européen et du Conseil</name>
  </names>
  <descriptions>
    <description lang="EN">Template 23 of the LS: Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL</description>
    <description lang="FR">Template 23 du SJ: Proposition de RÈGLEMENT DU PARLEMENT EUROPÉEN ET DU CONSEIL</description>
  </descriptions>
  <languages>
    <language lang="EN">English</language>
    <language lang="FR">Français</language>
  </languages>
</item>

<item type="TEMPLATE" id="PR-023" enabled="true" mandatory="true">
  <names>
    <name lang="EN">PR-023 - ....</name>
  </names>
  <descriptions>
    <description lang="EN">...</description>
  </descriptions>
</item>

<item type="DOCUMENT" id="ER-LP01" enabled="true" mandatory="true" default="true">
  <names>
    <name lang="EN">ER-LP01 - ....</name>
  </names>
  <descriptions>
    <description lang="EN">...</description>
  </descriptions>
</item>

<item type="DOCUMENT" id="BL-023" enabled="true" mandatory="true" default="true">
  <names>
    <name lang="EN">BL-023 - ....</name>
  </names>
  <descriptions>
    <description lang="EN">...</description>
  </descriptions>
</item>

<item type="DOCUMENT" id="SG-017" enabled="true" mandatory="false" default="false">
  <names>
    <name lang="EN">SG-017 - Annex to a proposal / to another proposal</name>
  </names>
  <descriptions>
    <description lang="EN">Annex to a proposal / to another proposal (COM/JOIN)</description>
  </descriptions>
</item>

<item type="DOCUMENT" id="FS-001" enabled="true" mandatory="true" default="true">
  <names>
    <name lang="EN">FS-001 - ....</name>
  </names>
  <descriptions>
    <description lang="EN">...</description>
  </descriptions>
</item>
</item>
```

There are some tags related with the text that appears in the template selection screen like names, descriptions and depending on the language selected some text or other will be shown.

Create new legislative document - Template selection (1/2)

Search by document type English

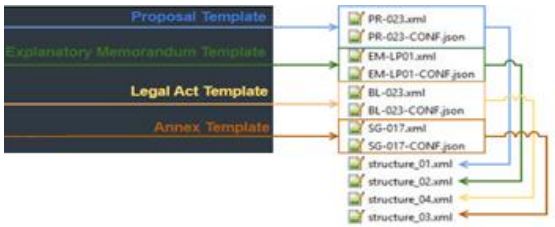
- Interinstitutional procedures - Law Initiative (COM/JOIN)
 - Legal acts
 - Ordinary legislative procedure
 - Proposal for a regulation
 - SJ-023 - Proposal for a Regulation of the European Parliament and of the Council**
 - Council legal acts
 - Proposal for a Council decision
 - SJ-019 - Proposal for a Council Decision

Template 23 of the LS: Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

Step 1 of 2:
Select the document type, language and name

Previous Next Create Cancel

These files **PR-023**, **EMP-LP01**, **BL-023** and **SG-017** are document and (sub)document templates in akn format and each of them can have associated a JSON configuration file.



In these JSON files some configuration like guidance or the structure file to be used to define all the elements that the document can contain can be added.

Sample BL-019-CONF.json

```
{
  "guidance": [
    {
      "targets": [...],
      "content": "<guidance id=\"guidance-cit-3\" class=\"guidance-green-block\"><span id=\"cit-3-guidance\">[where necessary]</span></guidance>"
    },
    {
      "targets": [...],
      "content": "<guidance id=\"guidance-cit-4\" class=\"guidance-green-block\"><span id=\"cit-4-guidance\">[where necessary]</span></guidance>"
    }
  ],
  //....
},
"alternatives": [
  {
    "name": "listOption1",
    "list": [
      {
        "index": "1",
        "content": "This Decision is addressed to the Member States."
      },
      {
        "index": "2",
        "content": "This Decision is addressed to the Member States in accordance with the Treaties."
      }
    ]
  },
  //...
]
},
"structure": [
  {
    "name": "structure_01"
  }
]
}
```

User Guidance

Alternatives

Structure file

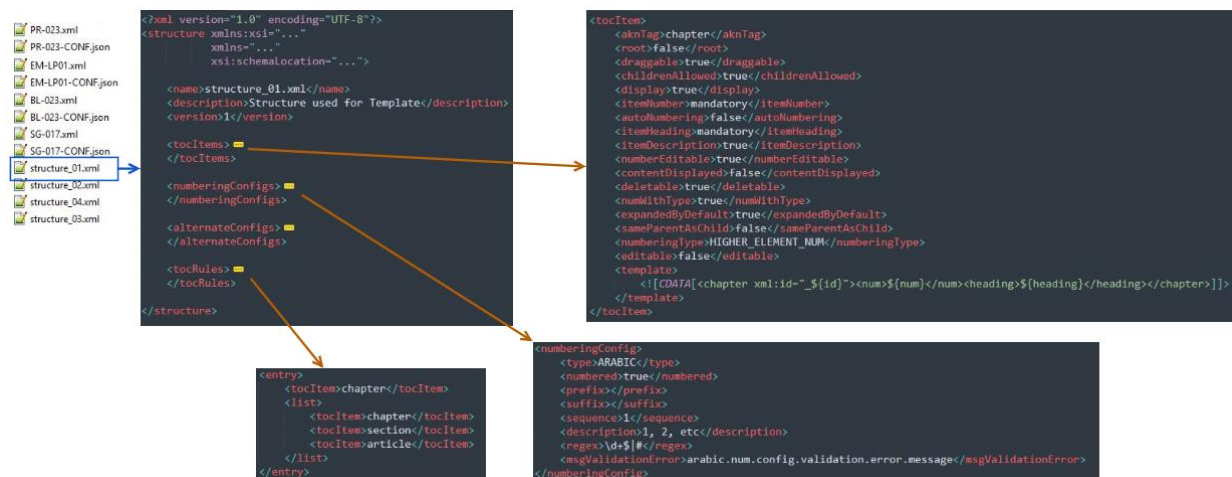
While LEOS comes by default with a set of document templates and configuration elements, based on your requirements, you can create your own document templates and configure according to your needs.

3.1. Templates Folder Structure

Under the path {AUGMENTED_LEOS}/leos/tools/repository/server/src/main/resources/leos/templates we can find 3 main paths:

- /templates/**N.X.Z** : where N.X.Z refer to specific release versions and contain and new templates or the modifications done to any existing template in the context of that specific release. This separation in release versions is relevant for the deployment process of LEOS within the European Commission under EdiT.
- /templates/**consolidated**: This folder contains the latest version of all existing templates currently configured in LEOS. It's only purpose is to serve as centralized reference point to facilitate the development of further templates
- /templates/**custom**: Will contain any custom templates to be loaded, as well as overwritten templates. This folder has precedence over the former templates folder, i.e., for example adding a modified catalog.xml file under this folder will make the document repository load it instead of the any other version of catalog.xml in the releases folder. This is useful to load any custom templates.

3.2. Structure File



The structure file further defines what elements shall be available in the in the document or subdocument it is being associated with, what structural and numbering rules should each of them obey to.

This schema is used to validate the structure files that configure all elements that can be used inside a document and that it is linked to a document template through template JSON configuration files. Some comments about this schema must be done:

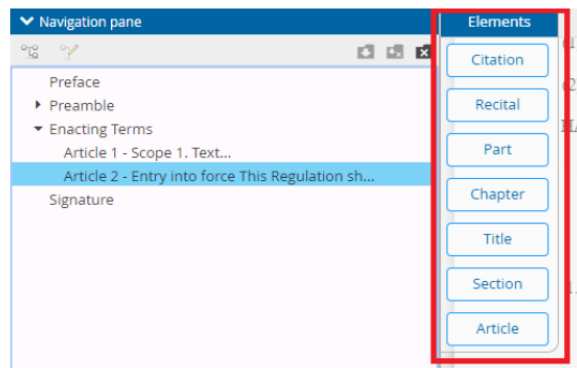
- If a new element must be added to a document, then this new element must be added to aknTag simpletype object.
- If a new numbering type must be added, then it must be added as a new numberingType object.

3.2. List of elements – TOC items list <tocItems>

TOC items list allows to define and configure a set of elements that can be used in the document or subdocument. For each element there's a set of properties to define its behaviour.

```
<tocItem>
  <aknTag>preface</aknTag>
  <root>true</root>
  <draggable>false</draggable>
  <childrenAllowed>false</childrenAllowed>
  <display>true</display>
  <itemNumber>none</itemNumber>
  <itemHeading>none</itemHeading>
  <itemDescription>true</itemDescription>
  <numberEditable>false</numberEditable>
  <contentDisplayed>false</contentDisplayed>
  <deletable>false</deletable>
  <numWithType>false</numWithType>
  <expandedByDefault>true</expandedByDefault>
  <sameParentAsChild>false</sameParentAsChild>
  <numberingType>NONE</numberingType>
  <editable>false</editable>
</tocItem>
```

- **aknTag**: Element name. It should be added or should exists in the structure schema too (structure_1.xsd inside domain module).
- **root**: If element it is a root element or not. In LEOS code there's some actions that cannot be done over root elements.
- **draggable**: If element is draggable it will appear in the elements pane. User can drag and drop element in the table of content tree.



- **childrenAllowed**: Element can contain other elements.
- **display**: Element is showed in the table of content tree.
- **itemNumber**: Element contains <num> tag.
- **itemHeading**: Element contains <heading> tag.
- **itemDescription**: Element text (ex. Article) is shown in the table of content tree.
- **numberEditable**: Number is editable by the user.
- **contentDisplayed**: Content inside the element is showed in the TOC.
- **deletable**: Element can be deleted.
- **numWithType**: Inside the num tag the element name (ex. Article) is with the number.
- **expandedByDefault**: Element is showed expanded in the TOC.
- **sameParentAsChild**: Element can contain same elements.
- **numberingType**: Numbering type used for numbering the element.
- **editable**: Element is editable.
- **template**: XML structure used when element is created in the document. By default, current template is used if no template is configured.

3.3. List of numbering configurations <numberingConfig>

Through the numberingConfigs list a set of different numbering configuration can be defined. A concrete number configuration is applied to an element using the <numberingType> tag inside the tocItem.

For example, ARABIC numbering type.

```
<numberingConfig>
  <type>ARABIC-PARENTHESES</type>
  <prefix></prefix>
  <suffix></suffix>
  <sequence>1</sequence>
  <description>(1), (2), etc</description>
  <regex>\d+$|#</regex>
  <msgValidationError>arabic.num.config.validation.error.message</msgValidationError>
</numberingConfig>
```

- **type**: Numbering configuration name. It should be added or should exist in the structure schema too (structure_1.xsd inside domain module).
- **prefix**: If numbering should include some prefix.
- **suffix**: If numbering should include some suffix.
- **sequence**: Numbering sequence (1 by 1 in this example).
- **description**: Just a description.
- **regex**: Regular expression used to validate number introduced by the user.
- **msgValidationError**: Error validation message showed defined inside application message.properties file.

3.4. List of table of content rules – <tocRules>

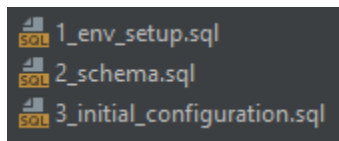
Allows to define which kind of elements can contain a particular element. For example, body element:

```
<entry>
  <tocItem>body</tocItem>
  <list>
    <tocItem>part</tocItem>
    <tocItem>title</tocItem>
    <tocItem>chapter</tocItem>
    <tocItem>section</tocItem>
    <tocItem>article</tocItem>
  </list>
</entry>
```

Body element can contain part, title, chapter, section and article elements.

3.5. Document repository Data Model

The Document repository stores all XML documents in a supporting database that needs to be configured and initialized prior of using LEOS. The steps to setup a MySQL database for the purpose can be found in {AUGMENTED_LEOS}/leos/tools/repository/web/src/etc/database/mysql/*



3.6. In-memory data configuration

Alternatively, and recommended to be used for testing purposes only, you can use the h2 in-memory database to load the templates.

To achieve this, besides adding the necessary AKN4EU template files as mentioned in **3.Document templates and element configuration**, a reference to the files should be added in the file {AUGMENTED_LEOS}/leos/tools/repository/server/src/main/resources/ **data-property-h2.sql**.

These reference implies an entry into :

- CONFIG table
- CONFIG_VERSION table

4. Template management

When running LEOS, it is required to execute the following four applications:

- **LEOS Core** - the main LEOS module where all the UI and Services reside
- **LEOS Annotate** - the module responsible for handling the comments and suggestions
- **LEOS User Repo** - to manage the users and roles
- **LEOS Doc Repo** - Document repository, where the templates are configured.

To configure a new Template, updates at the LEOS Doc Repo Level are needed:

1. Add the new files that help define the template:

- template XML,
- structure and
- -CONF.json files

in {AUGMENTED_LEOS}/leos/tools/repository/server/src/main/resources/leos/templates/custom

2. Make the necessary changes on the catalog.xml, and add the modified catalog.xml in the /custom folder as well. This is required to reflect the updates in the Create proposal wizard on the UI.

3. Re-launch the LEOS Doc Repo module so that the changes are loaded.

NOTE: It is not needed to relaunch the full LEOS Application every time changes to the templates are performed as this is managed on a separate module.